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MIL-HDBK- 540 (AV)

11 January 2023

DEPARTMENT OF DEFENSE HANDBOOK

COLD SPRAY REPAIR AND COATING OF AEROSPACE COMPONENTS

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FOREWORD

1. This handbook is approved for use by the U.S. Army Combat Capabilities Development Command (DEVCOM) Aviation & Missile Center (AvMC) and is available for use by U.S. Army Aviation and Missile Command (AMCOM), Program Executive Office (PEO) Aviation, and all Departments and Agencies of the Department of Defense (DoD).

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3. To meet the ongoing challenges of maintaining the Army's rotary-wing aircraft, it is necessary to introduce new technologies and processes to accomplish repairs that were previously considered impossible or uneconomical.

4. Cold Spray (CS) is a process that has the potential to allow a variety of new repairs. CS deposits of new metal in areas degraded or damaged, without the chemical baths required for electroplating or the high temperatures of fusion welding, thermal spray, or high-velocity oxygen-fuel (HVOF) deposition. However, as a line-of-sight process, CS is only applicable to exterior or highly accessible surfaces.

5. The CS process uses a high-velocity jet of particles to deposit material below the melting points of many engineering materials. This allows the CS process to apply deposits to a wide variety of substrates, while minimizing high temperature reactions characteristic of typical thermal spray processes, such as grain growth, oxidation, or phase changes. It is these characteristics of CS that makes it an attractive method for coatings, dimensional restoration, and the production of bulk materials or near-net shape parts.

6. The Army Research Laboratory (ARL) and other DOD agencies have been major proponents in developing and testing various CS materials and processes over the last 20 years. To date, non-structural repairs with aluminum CS material on magnesium have been approved, applied and are in use on Army rotorcraft components. More development continues for a variety of other CS materials, including Nickel-Carbon-Chrome and Tungsten Carbide-Cobalt-Nickel for chrome and nickel electroplating coating replacement. Potential substrates include magnesium; 6061, 7050, and 7075 aluminums; 4340 steel, 17-4 precipitation hardened and 316L stainless steels, and Nickel-Chrome-Molybdenum-Vanadium steels; Incoloy; and Ti-6Al-4V Titanium alloy.

7. This handbook is intended to provide actionable guidelines to entities looking to implement new CS repairs on aerospace components in compliance with the continued airworthiness requirements set forth in AR 70-62 and MIL-STD-3063 – Rotorcraft Structural Integrity Program, while maintaining the Corrosion Prevention and Control Plan (CPCP) and applicable Critical Safety Item (CSI) requirements. This handbook will also highlight the testing and qualification requirements that are unique to CS to aid in the introduction of these repairs.

8. Comments, suggestions, or questions on this handbook should be addressed to Director, U.S. Army Combat Capabilities Development Command (DEVCOM), Aviation & Missile Center (AvMC), ATTN: FCDD-AMR-SM, 5400 Fowler Road, Huntsville, AL 35898-5000, or e-mailed to usarmy.redstone.ccdc-avmc.mbx.avmc-xo@mail.mil. Since contact information can change, you may

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want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

9. Technical questions may be addressed to the following office: U.S. Army Combat Capabilities Development Command - Aviation and Missile Center, ATTN: FCDD-AMR-SM, 9035 Ocean Dr., Building 10, MS-21, Corpus Christi, TX 78419, Commercial (361) 961-3850 DSN 861-3850.

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1. SCOPE

1.1. Scope. This handbook embodies the general guidelines for assessing potential of implementing new CS repairs on metallic components and maintaining continuing airworthiness (AW) approval for Army aircraft. This handbook is for guidance only and cannot be cited as a requirement, contractual or otherwise.

2. APPLICABLE DOCUMENTS

2.1. General. The documents listed below are not necessarily all of the documents referenced herein, but are those needed to understand the information provided by this handbook.

2.2. Government documents.

2.2.1. Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein.

AEROSPACE DESIGN STANDARDS

ADS-13F-HDBK	Air Vehicle Materials and Processes
ADS-51-HDBK	Rotorcraft & Aircraft Qualification Handbook

DATA ITEM DESCRIPTIONS

DI-MISC-80711	Damage Repair Assessment
DI-MFFP-81403	Corrosion Prevention and Control Plan (CPCP)

MILITARY SPECIFICATIONS

MIL-DTL-32495	Powders for Cold Spray Deposition

MILITARY STANDARDS

MIL-STD-889	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapon Systems
MIL-STD-3021	Materials Deposition, Cold Spray
MIL-STD-3063	Rotorcraft Structural Integrity Program (RSIP)

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MIL-STD-7179	Finishes, Coatings, and Sealants, for the Protection of Aerospace Weapons Systems and Support Equipment

MILITARY HANDBOOKS

MIL-HDBK-6870	Nondestructive Inspection Program Requirements for Aircraft and Missile Materials and Parts
MIL-HDBK-896	Manufacturing Management Program Guide
	JACG Critical Safety Item Management Handbook

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2. Regulations. The following regulations form a part of this document to the extent specified herein.

ARMY REGULATIONS

AMACC	DEVCOM AvMC Army Military Airworthiness Certification Criteria
AMCOMR 702-7	Aviation Critical Safety Items, Critical Application Items, and New Source Testing Program Management – (Contains QE-STD-2)
AR 70-62	Airworthiness of Aircraft Systems

(Copies of the above documents are available online at https://armypubs.army.mil or from the U.S. Army Publishing Directorate, 9301 Chapek Road, Bldg.1458, Fort Belvoir, VA 22060)

2.3. Non-Government publications. The following documents form a part of this document to the extent specified herein.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.100M	Engineering Drawing Practices

(Copies of the above document may be obtained online at https://webstore.ansi.org or by contacting ASME International, Three Park Avenue, New York, New York 10016-5990)

BATTELLE MEMORIAL INSTITUTE

MMPDS	Metallic Materials Properties Development and Standardization Handbook

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(Copies of the above document may be obtained online at https://www.sae.org or contacting Battelle Memorial Institute at 505 King Avenue, Columbus, Ohio 43201)

2.4. Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1. Definitions

Aircraft. Any vehicle designed to be supported by air, being borne up either by the dynamic action of the air upon the surfaces of the vehicle, or by its own buoyancy. The term includes fixed and movable wing airplanes, helicopters, gliders, and airships, but excludes air-launched missiles. Aircraft is synonymous with air vehicle.

Airworthiness (AW). The property of an air system configuration to safely attain, sustain, and complete flight in accordance with approved usage limits.

Airworthiness Qualification. AW qualification is a progressive assessment process performed at the component, subsystem, and system levels to ensure that a system meets all applicable airworthiness requirements.

Airworthiness Release (AWR). An AWR is a technical document that provides operating instructions, procedures, limitations, inspections, and maintenance instructions necessary for safe flight and operation of an aircraft system, subsystem, component or allied equipment.

Airworthiness Qualification Plan (AQP). The AQP is a document which provides a comprehensive list of requirements for airworthiness qualification of a U.S. Army Aircraft System.

Airworthiness Qualification Specification (AQS). The AQS is prepared by the contractor or design authority and defines their obligation to conduct specific analyses, reviews, tests, surveys, and demonstrations to fulfill the requirements and objectives specified in the AQP.

Aviation Critical Safety Item (CSI). A part, an assembly, installation equipment, launch equipment, recovery equipment, or support equipment for an aircraft or aviation weapon system if the part, assembly, or equipment contains a characteristic any failure, malfunction, or absence of which could cause a catastrophic or critical failure resulting in the loss of or serious damage to the aircraft or weapon system, an unacceptable risk of personal injury or loss of life or an uncommanded engine shutdown that jeopardizes safety.

Cold Spray (CS). A materials deposition process in which small solid particles accelerated to high velocities to impact and adhere to a substrate. CS does not involve the melting or liquefying of the sprayed material or substrate, except as may occur from kinetic impact. A more detailed definition can be found in MIL-STD-3021

Continuing Airworthiness. Compliance with the processes (for example, aircraft maintenance, operator and maintainer training and certification, sourcing of spare parts, and configuration management) to ensure that, at any time in its life cycle, an aircraft complies with the technical conditions established at the issuance of the AWR or SAQ and is in a condition for safe operation.

Corrosion. Deterioration of a material or its properties due to the reaction of that material with its chemical environment.

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Criteria. Criteria is the functionality, property, or characteristic of interest.

Critical Characteristic (CC). Any feature throughout the life cycle of a critical safety item, such as dimension, tolerance, finish, material or assembly, manufacturing or inspection process, operation, field maintenance, or depot overhaul requirement, that if nonconforming, missing, or degraded may cause the failure or malfunction of the critical safety item. Items with a CC are CSIs.

Maintenance Engineering Call (MEC). The process and document to address deviations from prescribed maintenance procedures and/or inspection criteria, processes, and/or procedures in the appropriate maintenance technical manual (hard copy and electronic media, all versions) for the end item.

Maintenance Engineering Order (MEO). The MEO is a change to authorized maintenance practice(s) to ensure continuous AW and serves as a publication change document. It is used to update, correct, or change the current prescribed technical and/or maintenance procedures, inspection criteria, and processes in the appropriate technical manual (hard copy and electronic media, all versions) for the end item.

Maximum Material Condition (MMC). Condition of a feature which contains the maximum amount of material, that is, the smallest hole or largest pin, within the stated limits of size.

Manufacturing Stability. Condition in which materials, processes, joining methods, and structural concepts have matured to where consistent and repeatable quality, and predictable costs have been achieved to meet system production requirements. Also, process parameters and methods are understood, and robust and documented approaches for control of these factors, such as specifications, exist.

Method of Compliance (MoC). A logical or systematic means to demonstrate or verify compliance with AW criteria and standards. Method(s) of compliance will generate artifacts and objective evidence used in an AW authority's determination of AW. The MOC is the “proof” required by the AW evaluator.

Non-Destructive Inspection (NDI). The inspection of a structure or component in any manner that will not impair its future usefulness. The purpose of the inspection may be to detect flaws, measure geometric characteristics, determine material structure or composition, or it may characterize physical, electrical, or thermal properties without causing any changes in the part. Also called Non-Destructive Testing (NDT).

Repair. A repair is the process of restoring a component that no longer meets requirements to a condition that falls within the requirements.

Structure. Components that provide the strength, stiffness, and mechanical stability required for reacting, carrying, or transmitting loads or motions.

Structural Integrity. A condition in which none of the critical load-carrying elements or components of a structure fail in the service environment during a specified lifetime – for

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example, a structure's uncompromised ability to safely sustain the required load cycles at specified operating temperature extremes for the required duration.

3.2. Acronyms

ADS	Aeronautical Design Standard
AEC	Army Evaluation Center
AED	Aviation Engineering Directorate
ARL	Army Research Laboratories
AMACC	Army Military Airworthiness Certification Criteria
AMCOM	Aviation and Missile Command
AMRDEC	Aviation and Missile RDEC
AQP	Airworthiness Qualification Plan
AQS	Airworthiness Qualification Specification
AQSR	Airworthiness Qualification Substantiation Report
ASME	American Society of Mechanical Engineers
AvMC	Aviation & Missile Center (formerly AMRDEC)
AW	Airworthiness
AWIS	Airworthiness Impact Statement
AWR	Airworthiness Release
CC	Critical Characteristic
CEA	Cognizant Engineering Authority
CS	Cold Spray
DEVCOM	Combat Capabilities Development Command (formerly RDECOM)
DoD	Department of Defense
ESTCP	Environmental Security Technology Certification Program
FEA	Finite Element Analysis
FPP	Frozen Process Plan
FMECA	Failures Modes, Effects, and Criticality Analysis
HVOF	High Velocity Oxygen Fueled
JSSG	Joint Service Specification Guide
MEC	Maintenance Engineering Call
MEO	Maintenance Engineering Order
MMC	Maximum Material Condition
MoC	Method of Compliance
NAVAIR	Naval Air Systems Command
NDI	Non-Destructive Inspection
OEM	Original Equipment Manufacture
PDR	Preliminary Design Review
PAE	Preliminary Airworthiness Evaluation
PEO	Program Executive Office
RDEC	Research, Development and Engineering Center
RSIP	Rotorcraft Structural Integrity Program
SAE	Society of Automotive Engineers
SAQ	Statement of Airworthiness Qualification
SRD	Systems Readiness Directorate (formerly AED)

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TEMP	Test and Evaluation Master Plan
TEPP	Test and Evaluation Program Plan
UAS	Unmanned Aircraft System
UA	Unmanned Aircraft
WCN	Warnings, Cautions, Notes

4. GENERAL GUIDANCE

4.1. General. Any new or substitute materials, processes, and product forms proposal(s) for an aviation platform are required to demonstrate sufficient maturity for the proposed application in the five primary material design factors per MIL-STD-3063.

4.2. Types of Repairs. Base component material repairs fall into structural and non-structural repairs as determined by the responsible SRD Division: Propulsion or Structures and Materials. These SRD entities will determine the loading and stresses for the proposed CS repair area and component application. Both types of repairs will require review and approval by Structures and Materials in accordance with the established continuing AW process to assure that the repair procedures are complete and that process details are documented.

4.3. Other Considerations. Although not a technical requirement, programmatic and repair cost considerations should be understood before committing to a CS repair. Review and contrast the long-term need of a repair with that of the entire cost and timeline of repair development and qualification. For example, the CS deposition of aluminum on magnesium substrates took over 15 years to qualify and apply to a component for field installation and usage. Granted a portion of this time was advancing the Technology Readiness Level (TRL) to level 9; however, lessons learned from this history should not be overlooked.

4.4. Process Description. A detailed description of the CS process can be found in MIL-STD-3021.

4.5. CS Repair Checklist. A checklist of the necessary steps to evaluate the potential for performing a CS repair is provided in Appendix A.

4.6. Non-Structural Repairs. Any repair performed on components or items not identified as aircraft structure. Final determination which components are aircraft structure is at the discretion of the cognizant engineering authority. In the absence of such a determination, components should be presumed to be structural.

4.7. Repairs Within DMWR Limits. Repairs that are within DMWR limits are typically non-structural, cosmetic restoration repairs, e.g.: packing areas, sealing surfaces; corrosion/pitting; erosion/wear; or coating replacement. These repairs are performed on items that are not part of the aircraft structure.

4.8. Existing Metal Spray Repairs. For repairs within DMWR limits, if a similar repair process, such as Plasma Spray, HVOF, or Flame Spray is already approved on the part, the previous substantiation can provide the basis for an update repair using CS. In cases where such repairs were not feasible with these processes, the CS process may make the repair practical.

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4.9. New Repairs. If a repair process does not exist and the repair is within DMWR limits, leveraging a current approved CS repair on the same base material is the most direct path forward for approval.

4.10. Substantiation. For either scenario, prepare the repair procedure, process details, and historical substantiation for Structures and Materials review and approval.

4.11. Repairs Beyond DMWR Limits. For any non-structural repairs that exceed current DMWR limits or for substrates that do not have an approved CS repair, following the structural repair qualification steps is required.

4.12. Structural Repairs. Any repair performed on components or items identified as aircraft structure. Final determination which components are aircraft structure is at the discretion of the cognizant engineering authority. In the absence of such a determination, components should be presumed to be structural. Prior to submission to Propulsion or Structures and Materials, perform the basic analysis below to determine the CS repair potential.

4.13. Review Component Details. Identify the area of repair application, including 2D drawings and 3D model availability. Identify any available data on repair area substrate, and applicable CS repairs.

4.14. Review Component Interactions. Identify the system operating information: temperature range; environment-air, oil, grease, fluid, etc.; loading condition (unloaded, cyclic or static); fretting or sliding contact potential; impact of repair failure; repair failure detection; CSI status; impact to CSI Critical Characteristics; etc.

4.15. AW and System Requirements. Review the AQP and system level testing as required. From the information gathered, if CS repair failure would obviously negatively impact the system operation do not move forward with CS repair development. Document the analysis and conclusion.

4.16. Material Strength. Compare CS material and process options and resulting adhesion and tensile strengths with that of the application area base material strength. If the strength of the proposed CS material/process is less than the calculated required material strength, including safety factor, the part is not a CS repair candidate, document the analysis and conclusion. If the strength of the proposed CS material/process meets or exceeds the calculated required material strength, including safety factor, the part may be a CS repair candidate, move to the next step.

4.17. Additional Factors. Compare the CS repair area system operating information and perform an analysis into each aspect. Address each item and document the results.

4.18. Documentation. Regardless of the outcome, document the analysis and conclusion. Each documented analysis contains important lessons learned and builds the understanding of and confidence in the CS process.

4.19. Reporting and Approval. If the outcome is positive, prepare the proposed repair procedure, process details, material selection, historical substantiation if any, and

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information/analysis generated. Submit the CS repair plan and submission to Propulsion/Structures and Materials through the established MEC process.

5. DETAILED GUIDANCE

5.1. General. Throughout this section the term “CS” will include the pre- and post-processing as well as CS process itself. This will include any machining, cleaning, surface treatment, and NDI inherent to cold-spray application to the part. The CS application process for aluminum CS material on as-machined cast magnesium substrate for non-structural repairs was established by the Army Research Laboratory (ARL). The research investigation and documentation on the process and material properties are detailed in the Environmental Security Technology Certification Program (ESTCP) WP-200620 report. Other combinations of CS material and substrate should be validated through rigorous testing prior to repair development.

5.2. Material Selection. Given the broad range of possible options for CS media, material selection requires its own independent decision tree. For this section, assume a material selection has been made for the substrate, accounting for best possible combination of material characteristics for the component and its operational environment.

5.3. Technology Considerations. As an additive process, CS does not compromise the dimensions or geometry of components, as excess material can be subsequently removed by subtractive manufacturing methods. As the deposition process cannot always create a surface conforming to the required geometry, such as sharp interior or exterior corners, it is generally desirable to build the surface beyond the Maximum Material Condition (MMC) of the part and remove excess material to achieve required dimensions, tolerances, and surface finishes.

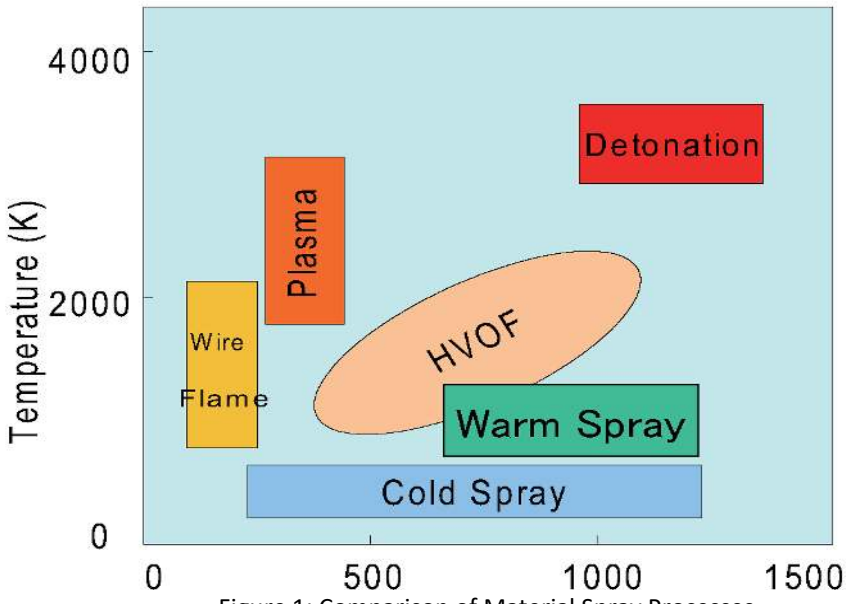


Figure 1: Comparison of Material Spray Processes

Warm Spraying - A novel coating process based on high-velocity impact of solid particles - S. Kuroda et al. - https://doi.org/10.1088/1468-6996/9/3/033002 - Creative Commons 4.0

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5.4. CS Advantages. As illustrated in Figure 1, the CS process differs from conventional thermal spray processes, primarily in the material deposition temperature, and thus offers several advantages:

a. Low process temperatures reduce oxidation, allow application of temperature sensitive materials, and reduce thermally induced microstructure changes.

b. Ability to achieve depositions with low (<5%) porosity and high adhesion

c. The strength and modulus values of the deposition can exceed those of wrought materials.

d. CS applied Aluminum and Aluminum-alloy can be harder than Aluminum wrought alloys due to spray-induced work hardening.

e. With a narrow spray plume, deposition location can be precisely controlled. Additional control can be obtained through masking and post-processing.

f. Most deposits are produced with compressive residual stress, which can result in favorable mechanical properties.

5.4.1. CS Limitations. Some of the limitations of the CS process and technology include:

a. CS depositions are limited by line of sight and process access requirements.

b. Deposition properties are sensitive to the spray's angle-of-impact.

c. Substrate preparation is critical to achieving adhesion and performance.

d. The stress state of a CS applied material may vary with thickness, geometry, and even within the deposit. Through analysis and testing of as-applied material may be required to ascertain the true state of the material.

e. Post-processing may be required to achieve the desired dimensions and surface finish.

f. Subsequent coating processes should be compatible with both the substrate and CS material.

5.5. Qualification Background. The methods for qualifying a CS repair will not differ from the established compliance process for continuing AW and will be the same as that used for qualifying any other repair to the same features of the same item. Qualification steps include focus on continuing AW at the component and system level, coating material, CS process, Non-Destructive Inspection (NDI), component processing, and production implementation.

5.6. AW Qualification. For new aircraft systems, AR 70-62 drives the AW process from Capability Requirement to Materiel Release. ADS and other standards support the AQP which feeds into the AQS, AW Qualification Substantiation Record (AQSR) or AW Release (AWR) and/or Statement of AW Qualification (SAQ), and then Material Release steps per AR 70-62

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figure 2-1. Be aware of the AQP and AQS analyses, reviews, tests, and demonstrations and address potential AW impacts from the CS component repair.

5.7. Continued AW. Once an aircraft is fielded, maintenance and repairs are controlled through continued AW. Per AR 70-62, any change that may affect the prescribed limits for continued AW, including any repair not authorized in the technical manual, such as a new CS repair, proceeds through the appropriate MEC, MEO, or DA 2028 technical requirement change process which incorporates the AR 70-62 AW authority approval process.

5.8. AW Substantiation. As a CS repair is developed, provide AW substantiation to ensure that damage repair procedures are developed in accordance with (IAW) MIL-STD-3063 sections: 5.4.3.4 Structural substantiation of damage repair limits and processes, 5.4.6 Technical manuals, and 6.3 Data requirements. A checklist of necessary steps can be found in Appendix B.

5.9. CSI Components. If part to be repaired is a CSI or part of a CSI component, identify and confirm affected Critical Characteristics (CCs) per AMCOMR 702-7 and QE-STD-2 and address any potential impacts. Ensure CSI and CCs are not deleteriously impacted and all frozen process planning is updated. Assess the potential for addition of CC(s) as well.

5.9.1. Approval Requirements. In accordance with the established requirements for continuing AW, each new CS repair MEO will need to be reviewed by Propulsion and Structures and Materials.

5.9.2. Cold Spray Structural Analysis Discussion. Analyze the structural effects of the CS repair and provide the results to the cognizant engineering authority for review. Such an analysis should be performed per recognized and established engineering practice, accounting for the conditions that the repaired components are required to endure. Perform the analysis in a variety of states and conditions to generate sufficient confidence in the performance envelope of the repair. General guidelines for performing the repair analysis are presented in Appendix C, though engineering judgement should be exercised to account for the circumstances of a particular repair. The analysis should be provided for continuing AW approval.

5.9.3. Potential Testing. In addition to the testing required by MIL-STD-3021 to qualify the CS process, the following tests have previously been used to the CS process. For future CS applications, this list can be used as a starting point for substantiating the repair. It may be necessary to perform certain tests under a variety of conditions, such as elevated or depressed temperatures, or cyclic conditions, as dictated by the operational environment of the component to be repaired.

a. Adhesion -

    1. Adhesion/Bond Strength Test (ASTM C633)

    2. Triple Lug Shear Testing (MIL J24445A)

    3. Impact/Drop (ASTM D2794, ASTM D5420)

b. Residual Stress -

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1. Almen Arc Measurement (SAE J442 Almen N)
2. Residual Stress Examination with X-ray Diffraction (XRD)

c. Fatigue -

1. Axial Fatigue (ASTM E466)
2. Fretting Fatigue
3. Subsequent Fractographic evaluation
4. RR Moore Rotating Beam Fatigue
5. Static Fatigue (slow strain rate)

d. Corrosion Testing -

1. Acidified Salt Fog (ASTM G85 method 4)
2. Crevice Corrosion (ASTM G78)
3. Filiform Corrosion (ASTM D2803)
4. Beach Exposure Testing - Kennedy Space Center or approved equivalent
5. Neutral Salt Fog Corrosion (ASTM B117)
6. Galvanic Stack-up (ASTM G71)
7. Mass Loss (GM9540)

e. Bulk Material Properties -

1. Hardness (ASTM E384 Vickers)
2. Metallography (ASTM E3, ASTM E112, ASTM E407)
3. Tensile (ASTM E8)

f. Wear -

1. Falex Block-on-ring (ASTMG77)
2. Tabor Abrasion (ASTM F1978)

g. Other Testing -

1. Pre-load Relaxation

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2. Recoatability (PATTI Adhesion and Bend Test)
3. Rosan Insert Test
4. Thermal Effects on Rosan Fittings
5. Eddy current (ASTM E1004)
6. Bend testing (MIL-S-13165C)

5.10. Cold Spray Material. The properties of a CS deposition are a combination of the CS material and CS application process. Refer to relevant scientific literature and testing, as well as MIL-STD-3021 and MIL-DTL-32495, to evaluate the applicability of specific materials and process combinations to the required repair.

5.10.1. Material Qualification. All material used in a CS process should be qualified to a relevant specification such as MIL-DTL-32495.

5.10.2. Material Testing. Testing may be required to determine as-sprayed CS material properties, which should be used to refine and update the repair analysis.

5.11. Cold Spray Application Process. All aspects of the Process, Material, Equipment, and Operators are to be qualified per MIL-STD-3021. The CS application process will vary for different CS materials, substrates, and part geometry. This section covers general items to be addressed, but each application is likely to require unique engineering judgement.

5.11.1. Substrate Preparation. Pre-Processing, including preparation of the CS application area by machining, surface finish, and cleanliness. Requirements for removal of oxides, surface preparations, and cleaning should be detailed.

5.11.2. CS Application. Establish and control process materials, parameters, and methods via specifications, standards, and manufacturing instructions.

5.11.3. Finishing. Post-Processing, including finishing machining operations, surface finish, and coating application.

5.11.4. Manufacturing Stability. All details of the CS process should achieve Manufacturing Stability prior to implementation. Manufacturing stability is key to establishing a CS frozen process plan (FPP).

5.12. NDI Requirements. Testing and qualification of selected NDI methods may be required per MIL-HDBK-6870. These may be required both where the CS repair is perform and where other NDI procedures may be effected by the repair.

5.12.1. Applicable NDI Methods. Potential NDI methods include fluorescent penetrant, magnetic particle, eddy current, ultrasonic, radiographic, and others. NDI experts should be consulted for NDI selection or development for CS repairs, as CS is a relatively new technology.

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5.12.2. CS Materials and NDI. While a properly applied CS material results in less than 5% porosity (MIL-STD-3021, section 5.2.3), this porosity may cause certain NDI methods, such as Ultrasonic or X-Ray inspection, to detect false failure indications.

5.12.3. NDI Requirements. The following should be the minimum requirements for addressing the NDI needs of the CS repair.

a. Address NDI of the deposition and at the substrate interface. Ensuring the integrity of the repair should be part of the manufacturing quality control procedures, but continuing verification in the field via NDI may be required.

b. Address NDI at the component level. Determine if the application of the CS repair interferes with the component's required NDI processes, including NDI for other repairs.

c. Address NDI at the system level. Determine if the application of the CS repair interferes with the system's required NDI processes, including NDI for other repairs.

5.12.4. NDI References. Reference MIL-STD-3063 sections 5.1.7, 5.1.9, 5.3.7, 5.4.3, 5.4.5, 5.4.6 and 6.3, using MIL-HDBK-6870 as a guide.

5.13. Corrosion Considerations. Evaluate the component's corrosion susceptibility, both at the CS repair interface with the base material and the CS material's interaction with the operating environment. Evaluations should be performed per MIL-STD-3063, section 5.2.12 Corrosion assessment.

5.13.1. System Level Corrosion. At the system level, review and address deviations from and changes to the platform's Corrosion Prevention and Control Plan (CPCP). Control corrosion and permit no materials or functional degradation to be caused by the application of the CS repair.

5.13.2. Material Compatibility. Even when the materials used in a CS repair are similar to the substrate they are applied to, even minor differences in the galvanic potential between the substrate and repair material may cause previously corrosion-free parts to experience uncharacteristic corrosion behavior. Such behavior should be anticipated, and mitigation measures incorporated into the repair process. Coatings typically used to prevent or mitigate corrosion, such as anodizing, conversion coatings, plating, or paints, should be compatible with both the component substrate and CS repair material. The compatibility of the CS repair material with other components in physical or electrical contact, as well as with standard maintenance practices and chemicals, should also be assessed.

5.14. Component Qualification. Once the CS FPP is completed, NDI Plan completed, and the part is CS repaired, then component qualification is required. Component qualification is performed per ADS-51-HDBK.

5.14.1. Identify Requirements. Review applicable AQP and AQS requirements and determine if similar testing can be used for the CS repair qualification.

5.14.2. Qualification Process. CS repaired component qualification general steps:

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a. Part is CS repaired.

b. Production process QC testing as applicable.

c. Component Inspection (including NDI).

d. System Assessment

e. System Inspection

f. Integration into the AQP and AQS as required.

g. Obtain CS frozen planning approval from the cognizant engineering authority (CEA).

h. Obtain Component Repair Qualification Approval from CEA

5.14.3. System Assessment. Part is assembled into system and operated under specific conditions and for a defined duration. Operating conditions are application specific, i.e.: engine, drive train, rotor, hydraulic, airframe, etc. Examples of testing include but are not limited to:

a. Low or high cycle fatigue testing

b. Reliability testing

c. Transmission or gearbox test cell

d. Engine run test and or accelerated test

e. Bench testing

f. Fielded airframe testing

5.14.4. System Inspection. Component system is visually inspected, NDI performed, and destructively tested as applicable. Upon completing system testing and inspection, the system is disassembled, and the repaired component subjected to visual, NDI, and destructive testing as applicable. All information is documented and entered into the qualification record.

5.15. Repair Process Implementation. Once the component CS repair has been qualified and approved for use, the repair process is implemented into the technical repair requirements.

5.15.1. Repair Implementation. The CS FPP new repair process is implemented into Maintenance Engineering Orders (MEOs) or DA Form 2028 revising affected Depot Maintenance Work Requirements (DMWR) or Technical Manuals (TM) respectively. Note that SRD Standard Operating Procedures (SOPs) control the review and approval for any technical change and incorporate the continuing AW process. The following are minimum requirements to document the repair process:

a. Part number(s) qualified for repair.

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b. Specific location(s) and dimensions of allowable repair area(s).

c. Document the damage repair limits: Highlight whether within existing DMWR limits or exceeding DMWR limits to a maximum depth.

d. MIL-STD-3021 compliance

e. CS material MIL-DTL-32495 compliance

f. CS FPP implementation.

g. Implementation in technical manuals per MIL-STD-3063.

5.15.2. Production Implementation. The repair entity or depot will need to qualify their CS repair process for each part IAW applicable CS technical repair requirements, resulting in site-specific CS frozen process planning, process standard and production quality control planning, IAW the requirements of MIL-STD-3021.

5.15.3. CS Process Planning. The cognizant engineering authority is responsible for approving all CS process planning. The process standard will be reviewed by the cognizant engineering authority and should contain production quality control sampling planning as required by MIL-STD-3021 and applicable quality control documentation.

6. NOTES

6.1. Intended Use. This handbook is intended to provide actionable guidelines to entities looking to implement new CS repairs on aerospace components in compliance with the continued airworthiness requirements set forth in AR 70-62 and MIL-STD-3063 – Rotorcraft Structural Integrity Program, while maintaining the Corrosion Prevention and Control Plan (CPCP) and applicable Critical Safety Item (CSI) requirements. This handbook does not present or intend to present any materials or methods to bypass or shortcut these processes, but rather to identify what information, data, and analysis is necessary to qualify a CS repair, as well as identifying the special concerns that should be addressed. This handbook also highlights the testing and qualification requirements that are unique to CS to aid in the introduction of these repairs. Using the guidelines outlined here, application of Cold Spray to structural repair of Army aviation components may be practical.

6.2. Cold Spray Process Development. Through ESTCP WP-0620 - Environmental Security Technology Certification Program: CS for Repair of Magnesium Components, November 2011, the U.S. Army Research Laboratory has demonstrated and validated a CS process using aluminum and/or aluminum alloys as a cost-effective, environmentally acceptable technology to provide surface protection and a repair/rebuild methodology to a variety of magnesium alloy components for use on Army and Navy helicopters and fixed-wing aircraft. Test results from this program indicate that CS as a repair technology for the reclamation of magnesium aerospace components offers improved performance and permits the reclamation of material properties, dimensional restoration, and improved readiness as replacements would be less frequent.

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6.3. Cold Spray of Aluminum. ESTCP WP-0620 results established a baseline for aluminum CS coatings as a replacement for current depot level thermal spray repairs. The mechanical test results of the 6061 Aluminum, Helium-carried coating that ARL developed suggest there is potential beyond non-structural or cosmetic repairs using CS technology. ARL has generated preliminary data, which shows that as-cold sprayed 6061 Aluminum has a higher ultimate tensile strength, yield strength, and hardness than conventional wrought 6061-T6 Aluminum.

6.4. Existing Cold Spray Qualifications. Evaluation and testing of Cold Spray repairs for Army aviation components dates to June 2005 and has been significantly building since 2012. Since then, CS repairs have been qualified on several magnesium components, notably the UH-60 Tail Gearbox and Outboard Retention Plate Assembly and the UH-60 Main Gearbox Input Module Assembly.

6.5. Information documents. The following documents are listed as references, but they are not specifically cited within the handbook.

AR 70-38	Research, Development, Test and Evaluation of Materiel for Extreme Climatic Conditions
AR 95-20	Contractor's Flight and Ground Operations
MIL-DTL-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapons Systems
MIL-DTL-31000	Technical Data Packages
MIL-HDBK-310	Global Climatic Data for Developing Military Products
MIL-HDBK-516C	Airworthiness Certification Criteria
MIL-HDBK-5400	General Guidelines for Electronic Equipment, Airborne
ESTCP WP-0620	Environmental Security Technology Certification Program: Cold Spray for Repair of Magnesium Components, November 2011
SERDP WP-2607	Cold Spray Coatings for Chromium and Nickel Plating Replacement
ESTCP WP19-5120	Cold Spray Coatings for Chrome and Nickel Plating Replacement
TM 1-1500-335-23	Technical Manual Nondestructive Inspection Methods, Basic Theory
TM 1-1500-366-23	Nondestructive Inspection General Procedures and Process Controls

(Copies of these documents are available online at https://assist.dla.mil, https://serdp-estcp.org, and https://www.dtic.mil)

6.6. Subject term (key word listing).

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Thermal Spray

Metal Coating

Structural Repair

Airworthiness Qualification

Additive Manufacturing

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APPENDIX A. COLD SPRAY REPAIR POTENTIAL CHECKLIST

A.1. Scope. The appendix provides a checklist of items and tasks to perform to determine if cold spray is a feasible repair method in a specific instance. This list is not exhaustive and users should use experience and judgement to identify additional information and analysis required to determine repair feasibility.

A.2. Gather Information

1. Identify the area of repair application

2. 2D drawings and 3D model availability

3. Structural or non-structural area

4. Propulsion or Materials and Structures component responsibility

5. Available data on repair area substrate

6. Available data on CS coating repairs

7. Identify the system operating information:

   7.1. Environment: Temperature Range, Air, Oil, Grease, Fluid, Etc.

   7.2. Loading: none, cyclic or static

   7.3. System Interactions: Galvanic coupling, Fretting, or Sliding

   7.4. Impact of coating failure

   7.5. Coating failure detection

   7.6. CSI status, CSI Critical Characteristic impact, etc.

8. Review the AQP and system level testing as required.

A.3. Analysis

1. From the information gathered, if CS repair failure would obviously negatively impact the system operation, do not move forward with CS repair development. Document the analysis and conclusion.

2. Compare CS material and process options and resulting adhesion and tensile strengths with that of the application area base material strength.

   2.1. If the strength of the proposed CS material/process is less than the calculated required material strength, including safety factors, the part is not a CS repair candidate. Document the analysis and conclusion.

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2.2. If the strength of the proposed CS material/process meets or exceeds the calculated required material strength, including safety factors, the part may be a CS repair candidate. Move to the next step.

3. Compare the CS repair area system operating information and perform a deep dive into all those aspects. Address each item and document the results.

   3.1. If the outcome is positive, prepare the proposed repair procedure, process details, material selection, historical substantiation if any, and information/analysis generated. Submit the CS repair plan and submission to Propulsion/Structures and Materials.

   3.2. If the outcome is negative, document the analysis and conclusion.

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APPENDIX B. COLD SPRAY REPAIR QUALIFICATION CHECKLIST

B.1. Scope. This appendix provides a checklist of the items and actions applicable to the qualification of a cold spray repair. This list is not exhaustive and all repairs are subject to the relevant airworthiness requirements and this list does not supersede those requirements. Users are encouraged to use engineering judgement to identify additional relevant repair qualification requirements.

B.2. AW

1. Initial: Review AR 70-62 AW Qualification Plan (AQP) and AW Qualification Specification (AQS) analyses, reviews, tests, and demonstrations and address potential AW impacts with the CS component repair.

2. Continuing AW: Per AR 70-62, assess any change that may affect the prescribed limits for continued AW for potential AW impact to the AW released system of record, including any repair not authorized in the technical manual, such as a new CS repair.

3. Structural Substantiation: Damage repair procedures are developed IAW MIL-STD-3063 sections 5.4.3.4 Structural substantiation of damage repair limits and processes, 5.4.6 Technical manuals, and 6.3 Data requirements.

4. Critical Safety Item: Identify and confirm CSI effected Critical Characteristics (CCs) per AMCOM Reg. 702-7 and QE-STD-2 and address any potential impacts.

B.3. Corrosion

1. Consideration of corrosion at the interface of the CS repair with the base material in addition to CS material interaction with the operating environment and other system components.

2. While the materials used in a CS repair may be similar to the substrate they are applied to, even minor differences in the galvanic potential between the substrate and repair material may cause previously corrosion-free parts to experience uncharacteristic corrosion behavior.

3. Reference MIL-STD-3063 - 5.2.12 Corrosion assessment.

4. At the system level, review and address Corrosion Prevention and Control Plan (CPCP) deviations and changes. Control corrosion and permit no materials or functional degradation.

B.4. Coating Material

1. Powder should be qualified an applicable standard such as MIL-DTL-32495

2. Note that coating properties are a combination of the CS material and CS application process.

3. Ensure that any new/substitute materials, processes, and product forms proposal(s) for an aviation platform demonstrate sufficient maturity for the proposed application (Technology Readiness Level 7+) in the five primary material design factors: Manufacturing Stability, Producibility, Characterization, Predictability and Supportability. Design factors definitions are included in MIL-STD-3063.

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4. Testing of as-sprayed coating material properties.

B.5. CS Application Process

1. Note that the CS application process will vary for all coating materials, substrates, and part geometry.

2. See MIL-STD-3021 for Process, Material, Equipment, and Operator qualification.

3. Establish and control process parameters and methods via specifications, standards, and manufacturing instructions.

4. Testing of coupons and scrap parts or equivalent bulk material, are required to substantiate the CS process and solidify the CS frozen process plan (FPP).

5. Residual stress or Almen strip deformation documentation

B.6. NDI

1. Note that because CS is a new technology, NDI experts should be consulted or employed for NDI selection or development for CS applications and repairs.

2. Note that CS coating typically has less than 5% porosity; NDI procedures may differ from inspection of billet, forgings, or other full density parts.

3. Note potential NDI methods include: FPI, MPI, Eddy Current, Ultrasonic, Radiology, Computed Aided Tomography Scan (CAT), Thermal, etc.

4. Address NDI needs and limitations.

5. Address NDI at the coating and substrate interface.

6. Address NDI at the component level.

7. Address NDI at the system level.

8. Reference MIL-STD-3063 sections 5.1.7, 5.1.9, 5.3.7, 5.4.3, 5.4.5, 5.4.6 and 6.3 using MIL-HDBK-6870 as a guide.

B.7. Component Qualification

1. Once the CS FPP is completed, NDI Plan completed, and the part is CS repaired, then component qualification required.

2. Per the ADS-51-HDBK, para 1-5.5 Sustainment Techniques: Qualification techniques that apply primarily during the operational and support phase of the life cycle of an item include testing and similarity. During the operational and support phase, qualification activities center primarily on changes to and improvement of an existing design, and similarity techniques are often applicable. When such techniques are not applicable, test and analysis may be used as appropriate. Another qualification technique is the assurance that repair items meet or exceed the criteria established by the original manufacturer.

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3. Review applicable AQP and AQS requirements and determine if similar testing could be employed for CS repair qualification.

4. CS component qualification general steps:

   4.1. Part is CS repaired.

   4.2. Production coupon testing as applicable.

   4.3. NDI part as applicable.

   4.4. Part is assembled into system and operated under specific conditions and duration. Operating conditions are application specific, i.e.: engine, drive train, rotor, hydraulic, airframe, etc. Examples of testing include but are not limited to:

      4.4.1. Low or high cycle fatigue testing

      4.4.2. Reliability test

      4.4.3. Transmission test cell

      4.4.4. Engine test run and or accelerated test

      4.4.5. Bench testing

      4.4.6. Fielded airframe test

   4.5. Part is dis-assembled from system, visually inspected, NDI as applicable, and destructively tested as applicable.

   4.6. Component system is visually inspected, NDI as applicable, and destructively tested as applicable.

   4.7. All information is documented into a qualification record.

   4.8. Integration into the AQP and AQS as required.

   4.9. Component Repair Qualification Approval and or CS frozen planning approval from the cognizant engineering authority.

B.8. Component Repair Process

1. Once the component CS repair has been qualified and approved for use, the repair process is implemented into the technical repair requirements.

2. The CS FPP is implemented into Maintenance Engineering Orders (MEOs) and or DA Form 2028 revising affected Depot Maintenance Work Requirements (DMWR) Technical Manuals (TM) respectively with the new repair process. Note that SRD SOPs control the review and approval for any technical change and incorporate the continuing AW process.

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3. MIL-STD-3063 - 5.4.6 Technical manuals. The rotorcraft technical manuals support Rotorcraft Structural Integrity Program (RSIP) by documenting structural operational limitations and restrictions, maintenance requirements, etc. as applicable.

4. Part numbers qualified for repair.

5. Specific location of allowable repair area (dimensions of size, depth, and location).

6. Document the damage repair limits: Highlight whether within existing DMWR limits or exceeding DMWR limits to a maximum depth.

7. CS material selection.

8. CS FPP implementation.

9. MIL-STD-3021 compliance.

B.9. Production Implementation

The repair entity or depot will qualify their CS repair process for each part IAW applicable CS technical repair requirements, resulting in site specific CS frozen process planning, process standard and production quality control planning and approval from the cognizant engineering authority. The process standard will be reviewed by the cognizant engineering authority and should contain production quality control sampling planning in conjunction with applicable CS specifications.

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APPENDIX C. COLD SPRAY LOAD ANALYSIS

C.1. Scope. This appendix provides information on performing load analysis on an item that has received a cold spray repair. This analysis procedure is presented to provide a baseline load analysis methodology, as cold spray techniques and experience grow users should adapt the analysis to current best practices and airworthiness requirements.

C.2. Required Information:
Detailed Load Information and Part Geometry. Computer models are beneficial where FEA is to be applied. Additional details on surface finishes will be helpful for fatigue analysis.

Mechanical Properties: Per MIL-STD-3021, page 4, Section 3.12 Structural applications. For applications where the cold spray deposit is a significant load-bearing member or where failure or disbonds could cause significant damage or reduction in service life.

Bond Strength: Per MIL-STD-3021. Comparison of the bond strength versus the substrate and CS material tensile strength may be used to determine if the repaired component should be analyzed as a single bulk material or as composite or bi-metallic material.

Ductility (Reference C.4.1, page 39) – Cold spray exhibits decrease in ductility (compared to the base metal). For a constant stress: Brittle materials are characterized by high elastic modulus and low strain. Ductile materials have a low Elastic Modulus and change shape considerably (Ductile materials include mild steel, aluminum and some of its alloys, copper, magnesium, nickel, brass, bronze).

C.3. Analysis Guidelines:

C.3.1. Apply conventional methods at net (geometry restored after Cold Spray). Assuming that the restored member is of homogeneous material substantiate using the minimum mechanical properties of the base material and/or the cold spray. In the case that the resulting safety margin is negative it is possible to rework the analysis as a bi-metallic material.

C.3.2. For uniformly distributed stresses (typically an axially loaded member, a beam in bending, or joint) apply net geometry (after restoration) with margins written against the minimum ultimate mechanical properties of the base material and/or the cold spray.

C.3.2.1. Axial Loading/Beam Bending: Restored Net Section against F_{tu}

C.3.2.2. Bearing at a Hole: Restored bearing geometry against F_{bru}

C.3.2.3. Shear Tearout against F_{su}

C.3.3. Finite Element Modelled part: Safety Margins - detailed stress analysis of geometric complexity

C.3.3.1. Ductile Material – Margins per the Octagonal Shear Stress theory:

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C.3.3.1.2. Tension/Shear: F_{ty} against the generated von-Mises stress.

C.3.3.1.3. Compression/Shear: F_{cy} against the generated von-Mises stress.

C.3.3.2. Brittle Material – Margins per the Maximum Normal Stress theory

C.3.3.2.1. Tension/Shear: margins should be written as F_{tu} against the envelope tensile principal stress.

C.3.3.2.2. Compression/Shear: margins should be written as F_{cy} against the envelope compressive principal stress.

C.3.4. Fatigue (Reference C.4.2 page 364): Bombardment of (cold spray) particles is found to add compressive residual stress with peening effect at the interface of the deposit; however, cold spray deposits are typically lower in ductility than the base metal. In lieu of established S/N curves for typical repaired geometry (base metal with cold spray build-up) it is recommended that any fatigue analysis use the discrepant geometry and base metal.

C.4. References

C.4.1. H. Assadi*, H. Kreye, F. Gärtner, T. Klassen; Cold Spraying a materials perspective, Helmut Schmidt University, Institute of Materials Engineering, Hamburg, Germany

C.4.2. Christian A. Widenera,b*, Ozan C. Ozdemira, and Michael Carterа; Structural repair using cold spray technology for enhanced sustainability of high value assets, 15th Global Conference on Sustainable Manufacturing

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CONCLUDING MATERIAL

Custodians:
Army – AV

Preparing activity:
Army – AV
(Project 16GP-2023-002)

Review activities:
Army – MI

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